

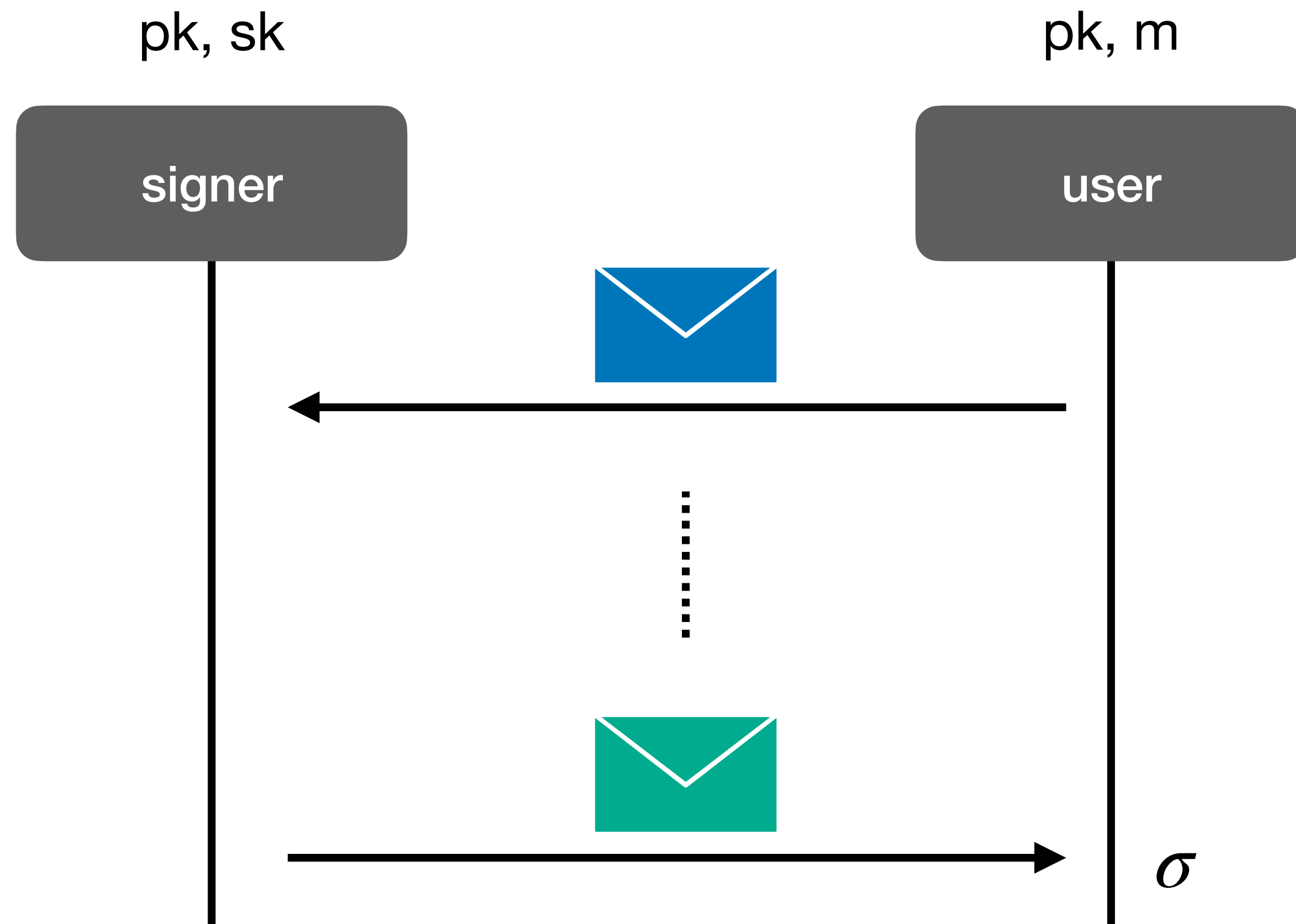
Pairing-free Blind Signatures in the ROM

Recent Progress and Techniques

Michael Reichle
ETH Zürich
Foundations of Cryptography Group



Blind Signatures



Correctness:

- honest signatures verify

Blindness:

- cannot link σ back to the signing sessions

One-more Unforgeability:

- user can obtain at most ℓ **signatures** from ℓ **sessions** with *distinct* messages

Goals

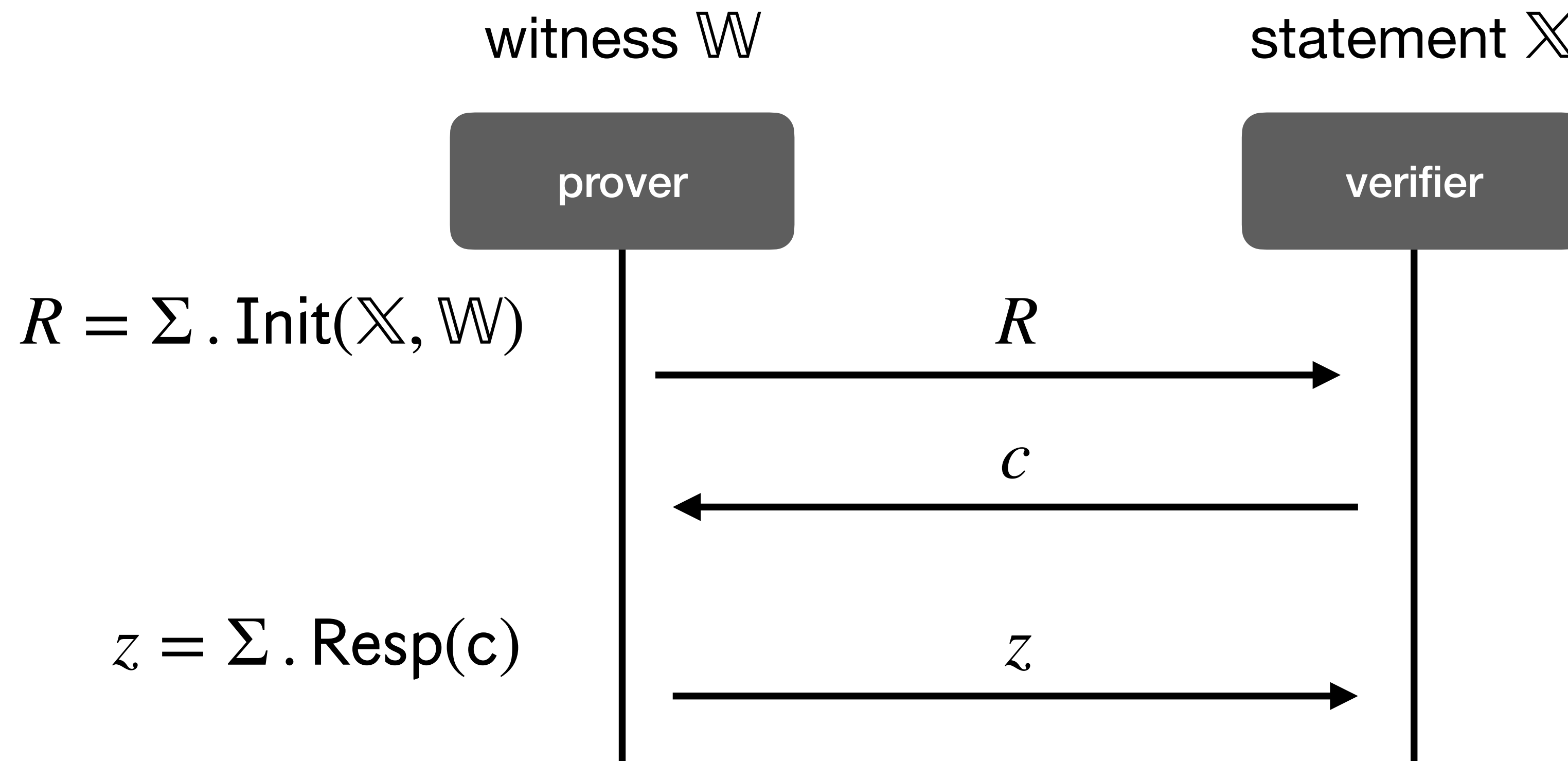
- **Pairing-free groups:**
 - group $\mathbb{G}$, prime order p , generator G
 - black-box construction
 - efficient
- **Assumptions:**
 - well-studied (DL, CDH, DDH)
- **Model:**
 - ROM (as hard to avoid)
 - avoid the AGM

Notation

Σ -protocols

Notation

relation $(\mathbb{X}, \mathbb{W}) \in \mathcal{R}$



Properties:

- correctness
- zero-knowledge
- soundness

Fiat-Shamir:

- choose $c = H(\text{pk}, R, m)$
- yields EUF-CMA signatures

Blind Schnorr

Insecure Example

! ignore blinding for the rest of the talk

DL-relation

$\text{pk} = \mathbb{X}, \text{sk} = \mathbb{W}$

signer

$\text{pk} = \mathbb{X}, m$

user

$R = \Sigma . \text{Init}(\mathbb{X}, \mathbb{W})$

R

ROS [BLLOR21]

c

blind R into $\tilde{R}$

$\tilde{c} = H(\text{pk}, \tilde{R}, m)$

blind $\tilde{c}$ into c

$z = \Sigma . \text{Resp}(c)$

z

blind z into $\tilde{z}$

$\sigma = (\tilde{R}, \tilde{c}, \tilde{z})$

Overview over Recent Progress

Pairing-free Setting, Black-box Constructions in the ROM

Overview [CTZ24]

Blind BLS

$$S = xH(m)$$

$$\text{pk} = X, \text{sk} = x$$

signer

$$\text{pk} = X, m$$

user

C



replace pairing-based verification of blind BLS
via FS-compiled Σ -protocol

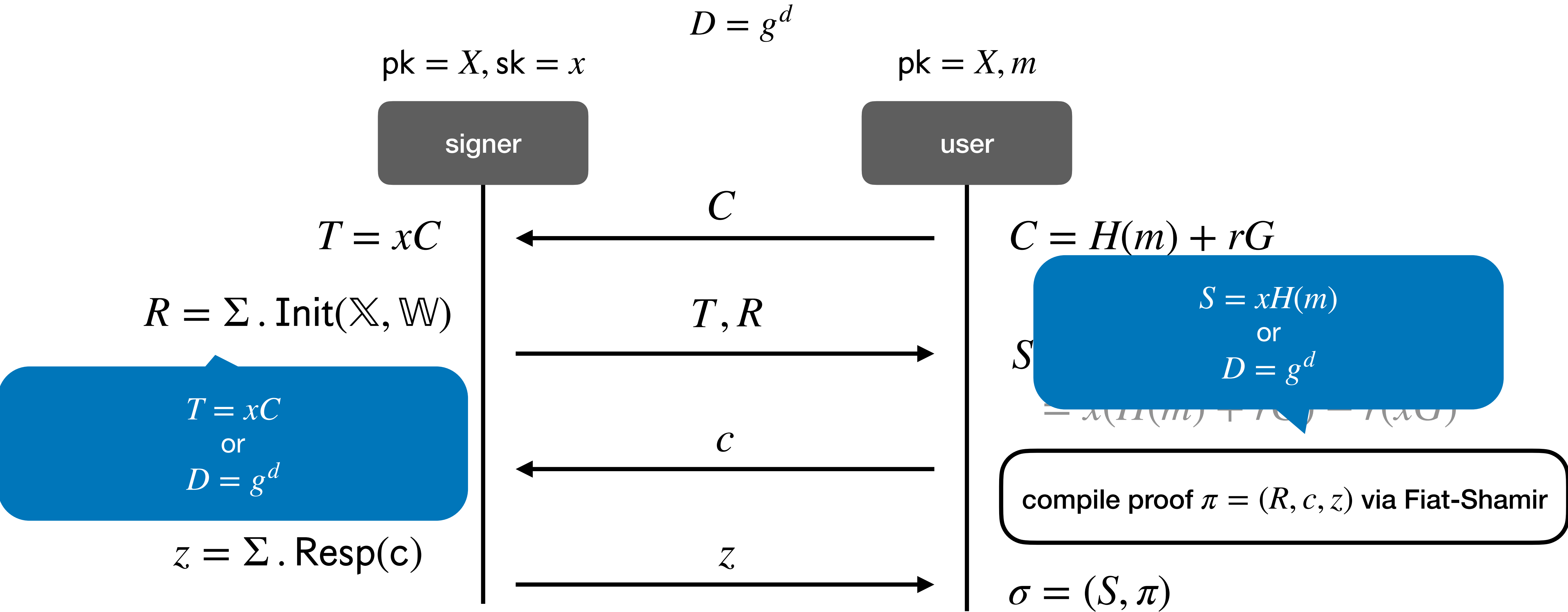


can be verified with a bilinear pairing
 $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ via $e(X, H(m)) = e(G, S)$

Overview [CTZ24]

Signing Protocol

$$S = xH(m)$$



Overview [CTZ24]

Security Analysis

- **Technique:**

- abort if adversary provides invalid BLS signature

→ justified by soundness of Σ -protocol (else, DL of D is found)

- employ DL-branch to sign

- use **OM-CDH** assumption to simulate BLS signature

→ $\ell + 1$ forgeries but only ℓ queries

Remark:

- **[CTZ24]** gives scheme based on CDH with overhead

Overview [KRW24]

Motivation

- **Starting Point:** [CTZ24]
- **Goal:**
 - remove reliance on one-more assumption
 - good efficiency
- **Idea:** employ [KRS23] instead of BLS

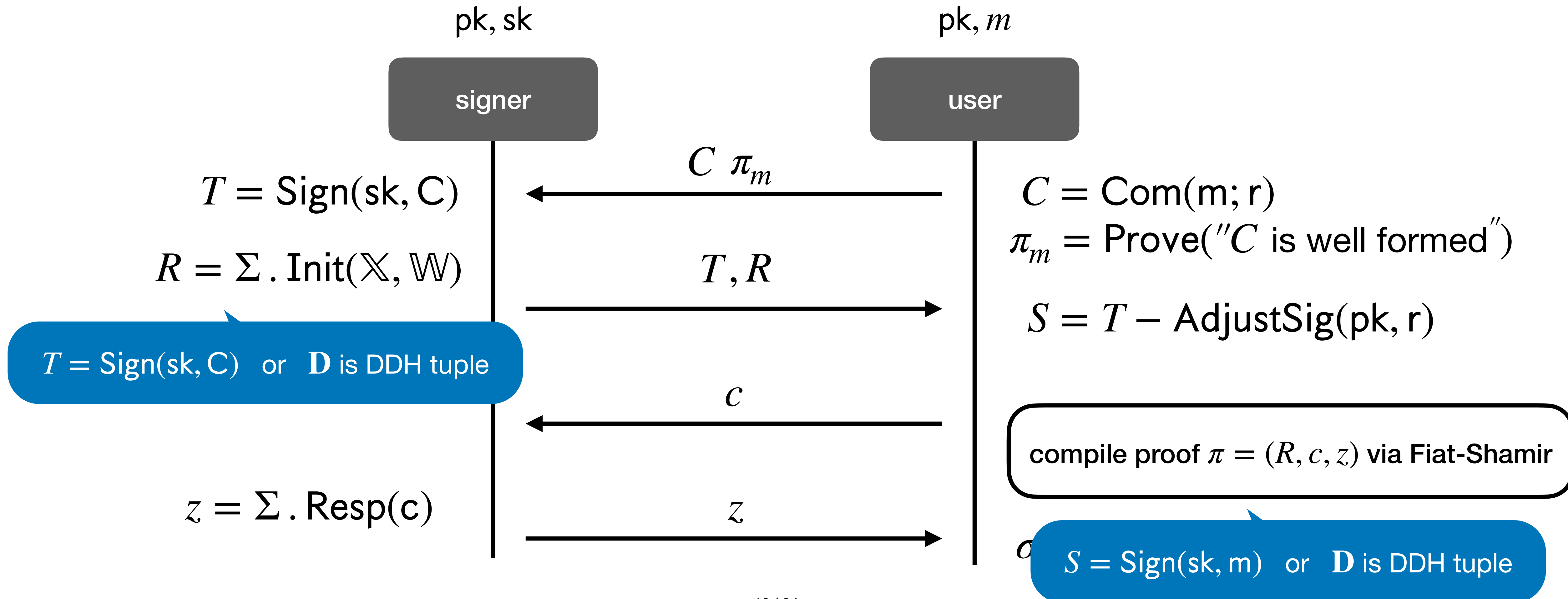


based on Boneh-Boyen IBE [BB04]

Overview [KRW24]

Signing Protocol

$$\mathbf{D} = (D_1, D_2, D_3)$$



Overview [KRW24]

Security Analysis

- **Technique:**

- abort if adversary provides invalid **[KRS23]** signature

→ justified by soundness of Σ -protocol (if D is not a DDH-tuple)

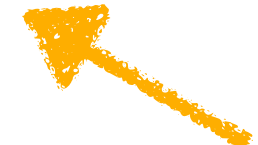
- employ DDH-branch to sign

- use **[BB04]** machinery to simulate **[KRS23]** signatures

a) puncture pk on some message m^*

b) force adversary to provide forgery for m^*

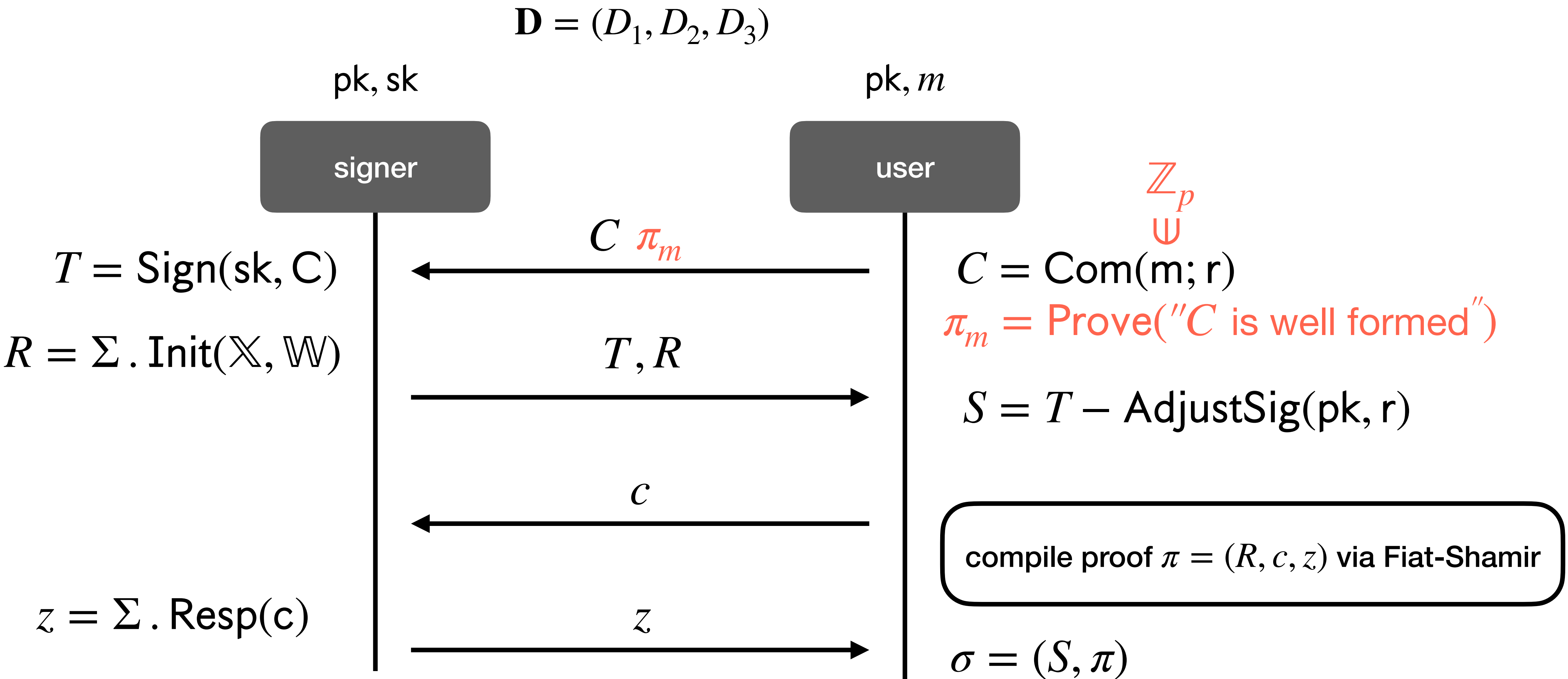
→ solve CDH



no rewinding required

Overview [KR25]

Motivation



Overview [KR25]

Techniques

- **Goal:** improve communication
- **Starting Point: [KRW24]**
 - remove reliance on NIZK Π for scalars in [KRW24]
- **Idea:**
 - employ tailored Σ -protocol
 - NIZK Π for group elements $\rightarrow$ less communication

Overview [KR25]

Tailored Trapdoor based on [BS02, CS03]

- **Idea:** craft tailored statement $\mathbb{X}$ for Fiat-Shamir such that
 - $\mathbb{X}$ can be punctured over $\mathbb{G}$ $\rightarrow$ message extracted from π_m is in $\mathbb{G}$
 - $\mathbb{X}$ is compact and linear $\rightarrow$ efficient blind issuance
- **Statement $\mathbb{X}$:** inequality of encrypted messages

$C := C^* - \text{Enc}(\text{pk}, M; 0)$ does not encrypt 0

Overview [KR25]

Signing Protocol

$$\mathbf{D} = (D_1, D_2, D_3 = d_1 D_2)$$

$$\text{pk} = (C^*, \mathbf{D}), \text{sk} = d_1$$

$$\text{pk}, M$$

signer

user

$$\mathbb{X} = C^* - C$$

$$R = \Sigma . \text{Init}(\mathbb{X}, \mathbb{W})$$

$C^* - C$ does not encrypt 0
or
 $\mathbf{D}$ is DDH tuple

$$z = \Sigma . \text{Resp}(c)$$

$$C \quad \pi_m$$

$$R$$

$$c$$

$$z$$

$$C = \text{Enc}(\text{pk}_{\text{rom}}, M; r)$$

$$\pi_m = \text{Prove}("C \text{ is well formed}")$$

compile proof $\pi = (R, c, z)$ via Fiat-Shamir

$C^* - \text{Enc}(\text{pk}, M; 0)$ does not encrypt 0
or
 $\mathbf{D}$ is DDH tuple

Overview [KR25]

Security Analysis

- **Technique:**

- employ inequality-branch to sign + switch D to random
- puncturing-based proof

a) puncture pk on some message M^* (i.e., encrypt M^* in C^*)

b) force adversary to provide forgery for M^*

wrong statement

→ forgery is proof that $C^* - \text{Enc}(pk, M^*; 0)$ does not encrypt 0

Overview [KLR25]

Motivation

- **Goal:** security proof under DL
- **Starting Point: [KR25]**
 - replace DDH-tuple D with random group element (as in [CTZ24])
 - replace ElGamal with Pedersen commitments
 - Statement $\mathbb{X}$: inequality of *committed* messages

$C := C^* - \text{Com}(\text{pp}, m; 0)$ does not commit to 0

Overview [KLR25]

Security Analysis

- **Technique:**
 - employ inequality-branch to sign
 - puncturing-based proof
 - a) puncture pk on some message M^* (i.e., commit to M^* in C^*)
 - b) force adversary to provide forgery for M^*
 - possible to forge as its Pedersen is only computationally binding

Overview [KLR25]

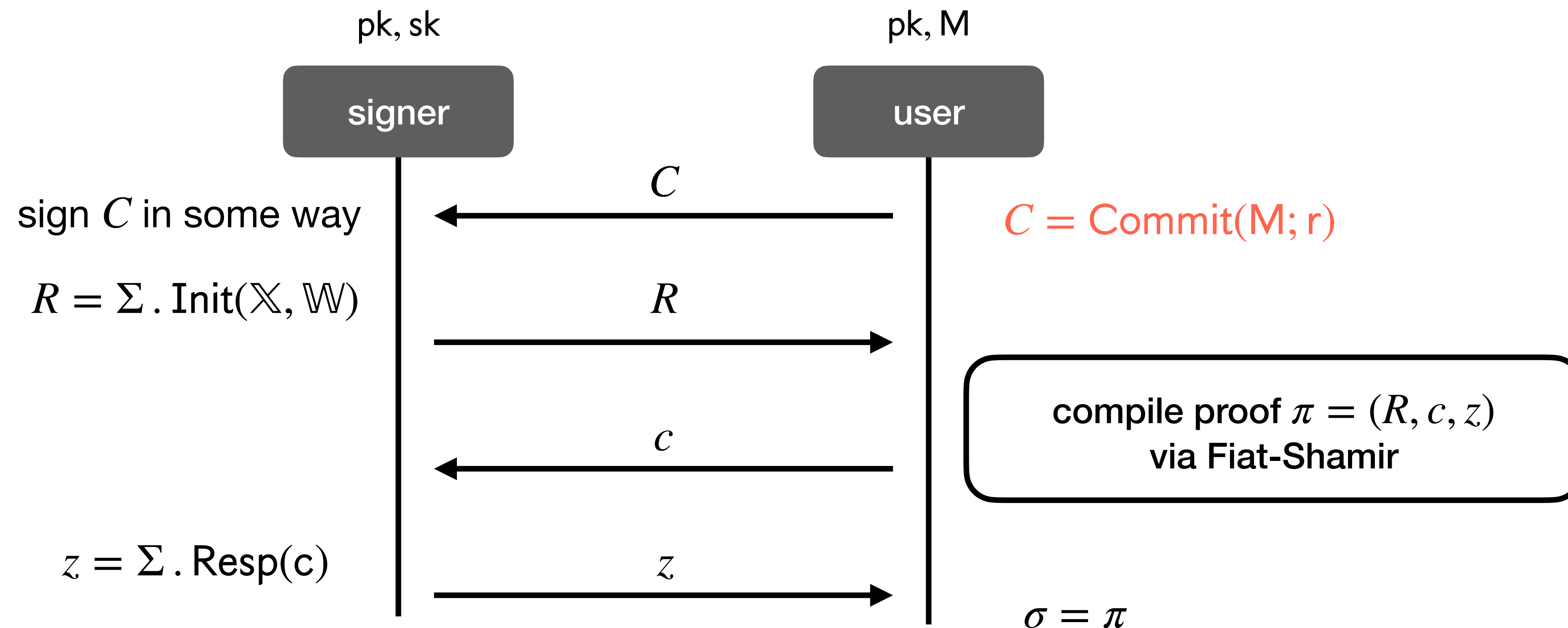
Security Analysis

- **Key Insight:**
 - after rewinding, we obtain a non-zero opening of $C^* - C$ by soundness
 - as we know opening of $C^* - C$ to zero, this breaks DL
- **Consequence:**
 - no witness-indistinguishability needed, any witness allows to solve DL
- **Note:**
 - very generic approach, can be instantiated from lattices

Overview [CRT26]

Motivation

- **Goal:** three-move protocol without AGM/GGM



Overview [CRT26]

Technique

- **Starting Point:** Okamoto-Schnorr [PS96]
- **Idea:**
 - tag the nonce in Okamoto-Schnorr with algebraic MAC (1st move by server)
 - show via NIZK that the algebraic MAC verifies
 - blind MAC and nonce, but commit to blinding factors and prove honest blinding
 - determines Okamoto-Schnorr witness *before* rewinding
- **Concurrent Work:** similar approach [Chen26]

Open Problems

- **Blind Signatures**

- 3-move protocol in ROM from DL
- more compact DL-based protocol
- other algebraic settings ([KLR25]: inefficient scheme from lattices)
- proofs for classical schemes / AGM-based schemes